

# Notes: Bongaerts, Cremers, and Goetzmann (JF, 2012)

## Tiebreaker: Certification and Multiple Credit Ratings

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# 1 Big picture

- **Question.** Why do corporate bonds often have multiple credit ratings, and what economic role is played by the marginal third rating agency?
- **Setting.** The paper studies corporate bond issues that are almost always rated by Moody's and S&P, and asks what the additional Fitch rating does in the 2000–2008 U.S. corporate bond market.
- **Three candidate mechanisms.** Multiple ratings may arise because: (i) the third CRA produces additional information; (ii) issuers shop for optimistic ratings; or (iii) the third CRA provides regulatory certification, especially at the investment-grade/high-yield (IG/HY) boundary.
- **Core institutional rule.** With two ratings, regulation and investment mandates often use the lower rating; with three ratings, they often use the middle rating. Therefore, if Moody's and S&P straddle the HY–IG boundary, Fitch can be a literal *tiebreaker*.
- **Main result.** The evidence supports regulatory certification much more strongly than information production or broad rating shopping.
- **Price result.** Fitch ratings generally do not add broad information to credit spreads once Moody's and S&P are controlled for, but they matter economically when they can change the regulatory IG/HY classification.
- **Selection result.** The probability of adding Fitch is higher when Moody's and S&P disagree around the HY–IG boundary, exactly where Fitch can move the issue into IG status under the median-rating rule.
- **Default prediction result.** Moody's and S&P dominate Fitch in default prediction; Fitch does not contribute much additional default-relevant information.
- **Liquidity result.** The regulatory classification effect is also visible in turnover: issues that Fitch moves into IG have stronger liquidity than otherwise similar boundary issues.
- **Economic takeaway.** In this market, the marginal rating is valuable mainly because regulation and clientele constraints attach discontinuous value to crossing the IG/HY boundary, not because the third rating broadly reveals new credit information.

# 2 Incremental contribution of the paper

- **From ratings as information to ratings as certification.** Prior work often treated ratings as signals of default risk or studied rating shopping. This paper adds a clean empirical distinction between information production, shopping, and regulatory certification.
- **Exploiting a mechanical institutional rule.** The paper's key insight is that the *same* Fitch rating can have different value depending on whether Moody's and S&P are on opposite sides of the HY–IG boundary. This creates a sharp test of certification.
- **Marginal rating rather than average rating.** Instead of asking whether ratings matter in general, the paper studies the marginal contribution of a third rating conditional on the first two ratings. This is the economically relevant decision for an issuer contemplating whether to pay for another rating.
- **Multiple empirical margins.** The paper triangulates the mechanism using (i) rating differences, (ii) spread reactions to rating changes/additions, (iii) time-to-Fitch hazard models, (iv) default-prediction tests, (v) rating-transition tests, and (vi) liquidity/turnover regressions.
- **Regulatory-arbitrage interpretation.** The paper clarifies why an optimistic CRA can be valuable even if its rating is not more informative: under the middle-rating rule, a favorable third rating can relax regulatory constraints and bring in investors who cannot hold high-yield debt.
- **Broader contribution.** The paper shows that market institutions can change the function of information intermediaries. Ratings are not only beliefs about credit risk; they are also inputs into regulation, contracting, index inclusion, and investor mandates.

## 3 Data and measurement

### 3.1 Bond and rating data

- The main sample uses quarterly bond-level observations from 2000 to 2008.
- Ratings data come from the Fixed Income Securities Database (FISD). The paper focuses on the three major CRAs: Moody's, S&P, and Fitch.
- The empirical sample generally requires bonds to have both Moody's and S&P ratings. Fitch is therefore interpreted as the marginal or third rating.
- Credit spreads are constructed from TRACE transaction data and are analyzed for a sample of bonds with observed prices and issue characteristics.
- Equity and firm characteristics are obtained from CRSP and Compustat and used as controls for risk, size, leverage, volatility, and other firm-level attributes.

**Interpretation:** The requirement that bonds have both Moody's and S&P ratings transforms the empirical question into: what does Fitch add at the margin?

### 3.2 Rating scale and rating boundaries

- Ratings are converted to a numerical scale where lower numbers correspond to better ratings: AAA/Aaa equals 1, AA+/Aa1 equals 2, and so on.
- The HY-IG boundary is the key threshold: BBB-/Baa3 or better is investment grade, while BB+/Ba1 or worse is high yield.
- The variable *Fitch Could Break Tie* equals one when Moody's and S&P are on opposite sides of the HY-IG boundary.
- The variable *Fitch Makes IG* indicates that the Fitch rating causes the median of the three ratings to be investment grade.
- The variable *Fitch Denies IG* indicates that the Fitch rating fails to move the issue into investment grade.

**Interpretation:** The paper does not rely only on continuous variation in ratings. It exploits a discontinuity created by regulation and mandates.

### 3.3 Credit spread and liquidity variables

- Credit spreads are measured in basis points over the relevant benchmark and summarized by rating category.
- Credit spread changes are measured around rating upgrades, downgrades, and Fitch additions.
- Bond turnover is measured as quarterly trading volume divided by the total value of outstanding bonds.
- Turnover is important because a rating that moves a bond into the IG category can expand the eligible investor base and increase liquidity.

**Interpretation:** If Fitch matters through certification, the effect should show up not only in spreads but also in trading activity around the boundary.

### 3.4 Uncertainty and risk controls

- Rating disagreement and *MSP Rating Dispersion* measure the distance between Moody's and S&P ratings.
- Analyst dispersion measures disagreement in equity analyst forecasts.
- Idiosyncratic volatility measures firm-specific equity risk.
- Controls include issue size, redeemability, duration, convexity, total assets, leverage, return on assets, R&D, PPE, beta, and turnover.

**Interpretation:** These variables help distinguish information-production or rating-shopping motives from the certification motive.

## 4 Hypotheses and empirical predictions

### 4.1 Information production

- If Fitch produces useful incremental information, adding a Fitch rating should reduce uncertainty and lower credit spreads even away from the HY–IG boundary.
- Fitch rating changes should affect spreads similarly to Moody's and S&P rating changes after controlling for other ratings.
- Fitch ratings should improve default prediction when added to Moody's and S&P ratings.

**Interpretation:** Information production predicts a broad, not boundary-specific, role for Fitch.

### 4.2 Rating shopping

- If issuers shop for ratings, Fitch additions should be optimistic relative to existing ratings.
- Optimism should be most valuable when a better rating lowers spreads or changes regulatory classification.
- Rating shopping can be strongest near the HY–IG boundary, but pure rating shopping should also imply price benefits from favorable ratings away from the boundary if investors take them seriously.

**Interpretation:** Rating shopping focuses on issuer selection into favorable public ratings; it can mimic certification around the boundary but has different implications away from the boundary.

### 4.3 Regulatory certification

- If ratings are valuable because they determine IG/HY eligibility, Fitch should matter most when Moody's and S&P straddle the HY–IG boundary.
- Fitch should be more likely to be added precisely where it can break the tie.
- Spreads and liquidity should improve when Fitch makes the bond investment grade.
- Fitch need not improve default prediction if its main role is classification rather than information.

**Interpretation:** This hypothesis predicts a localized effect at a regulatory discontinuity, not a universal information effect.

## 5 Empirical specifications and interpretation

### 5.1 Rating differences

$$RatingDifference_{i,t}^{A,B} = Rating_{i,t}^A - Rating_{i,t}^B, \quad (1)$$

where lower numerical ratings correspond to better ratings. A negative value means agency  $A$  is more optimistic than agency  $B$ .

**Interpretation:** This test asks whether Fitch is systematically more optimistic than Moody's and S&P, a necessary pattern for either shopping or certification to matter.

### 5.2 Credit spread change regressions

$$\Delta Spread_{i,t} = \alpha + \beta_1 Upgrade_{i,t}^M + \beta_2 Downgrade_{i,t}^M + \beta_3 Upgrade_{i,t}^S + \beta_4 Downgrade_{i,t}^S + \beta_5 Upgrade_{i,t}^F + \beta_6 Downgrade_{i,t}^F + \Gamma X_{i,t} + \varepsilon_{i,t} \quad (2)$$

The paper also includes interactions for Fitch rating changes that break the HY-IG tie.

**Interpretation:** If Fitch is informative, Fitch upgrades and downgrades should move spreads broadly. If Fitch certifies, its effect should be concentrated when it changes IG/HY classification.

### 5.3 Fitch addition regressions

$$\Delta Spread_{i,t} = \alpha + \beta FitchAdded_{i,t} + \gamma BoundaryEffect_{i,t} + \Gamma X_{i,t} + \varepsilon_{i,t}. \quad (3)$$

**Interpretation:** The addition test asks whether the act of adding Fitch is associated with price changes, and whether the sign and size depend on Fitch making or denying investment-grade status.

### 5.4 Cox hazard model for adding Fitch

$$h_i(t) = h_0(t) \exp\{\beta_1 FitchCouldBreakTie_{i,t} + \beta_2 RatingDispersion_{i,t} + \beta_3 Uncertainty_{i,t} + \Gamma X_{i,t}\}. \quad (4)$$

**Interpretation:** The Cox model treats the Fitch addition as an event. A positive coefficient on *Fitch Could Break Tie* is direct evidence that issuers seek Fitch when the third rating has regulatory value.

### 5.5 Default prediction regressions

$$Pr(Default_{i,t+h} = 1) = \Lambda(\alpha + \beta_M MoodyRating_{i,t} + \beta_S SPRating_{i,t} + \beta_F FitchRating_{i,t} + \Gamma X_{i,t}). \quad (5)$$

**Interpretation:** If Fitch adds information,  $\beta_F$  should be significant after controlling for Moody's and S&P and prediction accuracy should rise. The paper generally does not find this.

### 5.6 Turnover regressions

$$Turnover_{i,t} = \alpha_i + \tau_t + \beta_1 HY_{i,t} + \beta_2 FitchMakesIG_{i,t} + \beta_3 FitchDeniesIG_{i,t} + \Gamma X_{i,t} + u_{i,t}. \quad (6)$$

**Interpretation:** The turnover tests link certification to liquidity. If IG status increases the eligible investor base, Fitch-induced IG classification should increase turnover or prevent a liquidity drop.

## 6 Identification and empirical design

### 6.1 Source of identifying variation

- The paper uses within-bond and cross-bond variation in whether Fitch is present, whether Fitch is added, and whether Fitch can change the IG/HY classification.
- The strongest quasi-discontinuous variation comes from Moody's and S&P being on opposite sides of the HY-IG boundary.
- The same Fitch rating has much higher regulatory value when it is the median rating that determines IG/HY status.

**Interpretation:** The paper's design is not a randomized experiment; it is a set of mechanism tests that exploit institutional rules and multiple market outcomes.

### 6.2 Threats and mitigations

- **Selection into Fitch.** Issuers choose whether to obtain Fitch, so the paper models the selection using hazard regressions.
- **Credit-quality confounding.** Average Moody's/S&P ratings and firm controls absorb many credit-risk differences.
- **Information versus certification.** Default-prediction tests and spread tests away from the boundary help separate information from certification.
- **Liquidity channels.** Turnover regressions test whether the certification effect is also reflected in market trading.

**Interpretation:** The core credibility comes from observing a coherent pattern across rating levels, spreads, default prediction, Fitch-addition timing, and liquidity.

## 7 Tables and figures

### 7.1 Table I: Empirical Predictions + Figure 1: FISD Database Coverage by CRA

Read:

- Table I maps the three theories to six testable predictions. It is the paper’s roadmap.
- Information production predicts that agreeing ratings and uncertainty should matter broadly.
- Rating shopping predicts optimistic additional ratings and spread benefits if the market accepts them.
- Regulatory certification predicts a special effect when the third rating can move the issue into IG status.
- Figure 1 shows that after 2000 Moody’s and S&P coverage is very high, while Fitch covers a smaller but meaningful share of issues.

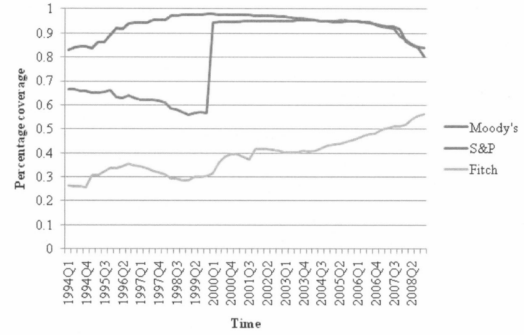
**Economic message:** The paper can focus on Fitch as the marginal rating because Moody’s and S&P are almost always present for large corporate bonds.

**Table I**  
**Empirical Predictions**

The various empirical predictions of the three hypotheses we consider are summarized in the table below, where “–” indicates that the implication is not supported and “+” means that it is supported.

| Reason for Multiple Ratings                                                                        | Information Production | Rating Shopping | Regulatory Certification |
|----------------------------------------------------------------------------------------------------|------------------------|-----------------|--------------------------|
| (i) Additional agreeing rating lowers credit spreads                                               | +                      | –               | –                        |
| (ii) Additional relatively optimistic rating lowers credit spreads (also away from HY–IG boundary) | +                      | +               | Only at HY–IG boundary   |
| (iii) Uncertainty uniformly increases # of ratings                                                 | +                      | +               | possible                 |
| (iv) Additional rating more likely if that could push the issue into IG classification             | possible               | possible        | +                        |
| (v) Additional rating more optimistic (especially around HY–IG boundary)                           | –                      | +               | Only for strategic CRA   |
| (vi) Additional rating associated with higher expected time variation in ratings                   | +                      | –               | +                        |

(a) Table I: Empirical Predictions



**Figure 1. FISD database coverage by CRA.** The figure plots the percentage of bonds in FISD covered by S&P, Moody’s, and Fitch. While FISD starts earlier, we use ratings starting in 2000, as S&P data appear to be incomplete prior to 2000, as this figure suggests.

(b) Figure 1: FISD Database Coverage by CRA

### 7.2 Table II: Summary Statistics for Credit Spreads Sample + Figure 2: Average Credit Spreads by Rating Category

Read:

- Table II shows 44,366 quarterly observations in the credit-spread sample with both Moody’s and S&P ratings.
- Fitch is present in about 68% of observations, and the average Fitch rating is more optimistic than the average Moody’s or S&P rating.
- The mean credit spread is about 173 basis points, but spreads vary greatly across the credit-quality distribution.
- Figure 2 shows the expected monotonic spread pattern: worse ratings carry higher spreads, and spreads rise sharply during the crisis period.

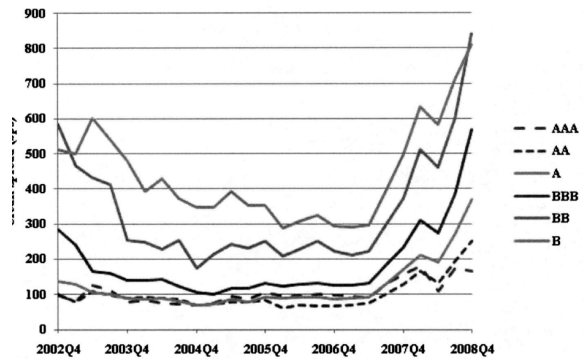
**Economic message:** The sample contains rich variation in rating levels, rating disagreement, Fitch presence, and spreads, allowing the paper to distinguish mechanisms.



Table II  
Summary Statistics for Credit Spreads Sample

The table presents summary statistics and a brief description of the sample of bond issues that have both a Moody's and an S&P rating in the quarterly credit spreads sample for 2002 to 2008.

| Variable                | N      | Mean   | Std. Dev. | Min     | Max    | Explanation                                             |
|-------------------------|--------|--------|-----------|---------|--------|---------------------------------------------------------|
| Fitch Could Break Tie   | 44,366 | 0.032  | 0.18      | 0       | 1      | Moody's and S&P on opposite sides of the HV-IG boundary |
| Fitch Rated             | 44,366 | 0.68   | 0.47      | 0       | 1      | Rated by Fitch                                          |
| Fitch Rating            | 44,366 | 5.04   | 4.17      | 0       | 16     | Fitch rating                                            |
| Moody's Rating          | 44,366 | 7.33   | 3.6       | 1       | 18     | Moody's rating                                          |
| S&P Rating              | 44,366 | 7.17   | 3.55      | 1       | 17     | S&P rating                                              |
| Fitch Makes IG          | 44,366 | 0.016  | 0.12      | 0       | 1      | Fitch pulls IG                                          |
| Fitch Denies IG         | 44,366 | 0.0066 | 0.081     | 0       | 1      | Fitch denies IG                                         |
| MSP Rating Dispersion   | 44,366 | 0.43   | 0.69      | 0       | 12     | Absolute value of MSP rating difference                 |
| Moody's Upgrade         | 44,366 | 0.027  | 0.16      | 0       | 1      | Moody's upgrade (common sample)                         |
| Moody's Downgrade       | 44,366 | 0.034  | 0.18      | 0       | 1      | Moody's downgrade (common sample)                       |
| S&P Upgrade             | 44,366 | 0.029  | 0.17      | 0       | 1      | S&P upgrade (common sample)                             |
| S&P Downgrade           | 44,366 | 0.038  | 0.19      | 0       | 1      | S&P downgrade (common sample)                           |
| Fitch Upgrade           | 44,366 | 0.017  | 0.13      | 0       | 1      | Fitch upgrade (common sample)                           |
| Fitch Downgrade         | 44,366 | 0.024  | 0.15      | 0       | 1      | Fitch downgrade (common sample)                         |
| Fitch Added, Better     | 44,366 | 0.0039 | 0.062     | 0       | 1      | Fitch added and < MSP                                   |
| Fitch Added, Equal      | 44,366 | 0.005  | 0.071     | 0       | 1      | Fitch added and = MSP                                   |
| Fitch Added, Worse      | 44,366 | 0.0003 | 0.017     | 0       | 1      | Fitch added and > MSP                                   |
| Credit Spread           | 44,366 | 172.71 | 141.72    | 0.12    | 999.86 | Credit spread                                           |
| Change in Credit Spread | 44,366 | 18.39  | 68.33     | -478.36 | 498.42 | Credit spread change                                    |
| Redeemable              | 44,366 | 0.69   | 0.46      | 0       | 1      | Callable                                                |
| Log of Offering Amount  | 44,366 | 12.03  | 1.88      | 0       | 15.42  | Log of offering amount                                  |
| Duration                | 44,366 | 6.32   | 3.72      | 1       | 18.91  | Duration                                                |
| Idios. Vol.             | 44,366 | 0.016  | 0.0084    | 0.0012  | 0.11   | Idiosyncratic stock volatility                          |
| Log of Total Assets     | 44,366 | 10.34  | 1.55      | 5.34    | 13.65  | Log of total assets                                     |
| Leverage                | 44,366 | 0.34   | 0.17      | 0       | 5.77   | Leverage                                                |
| ROA                     | 44,366 | 0.014  | 0.02      | -0.46   | 0.26   | ROA                                                     |
| PPE/Total Assets        | 44,196 | 0.36   | 0.24      | 0       | 0.95   | PPE/Total assets                                        |
| R&D/Total Assets        | 44,366 | 0.012  | 0.023     | 0       | 0.23   | R&D/Total assets                                        |
| R&D Missing             | 44,366 | 0.43   | 0.5       | 0       | 1      | R&D missing                                             |
| Analyst Dispersion      | 43,923 | 0.0033 | 0.013     | 0       | 1.1    | Analyst dispersion                                      |
| Beta                    | 44,366 | 0.95   | 0.39      | -0.25   | 4.21   | Equity beta                                             |
| Convexity               | 44,366 | 53.83  | 56.58     | 1       | 357.47 | Convexity                                               |
| Turnover                | 43,292 | 13.15  | 15.27     | 0.018   | 85.99  | Trading volume over offering amount, times 1K           |



(a) Table II: Summary Statistics for Credit Spreads Sample

(b) Figure 2: Average Credit Spreads by Rating Category

### 7.3 Table III: Average Rating Differences + Figure 3: Rating Differences across CRAs

Read:

- Table III shows Fitch is on average more optimistic than both Moody's and S&P. Negative differences mean the first agency gives a better rating.
- Fitch versus Moody's is around -0.538 notches for all bonds; Fitch versus S&P is around -0.397 notches.
- The Fitch optimism is strongest around some lower rating groups, where regulatory classification is especially important.
- Figure 3 visualizes the same pattern: Fitch bars are generally below zero relative to the other agencies, while Moody's versus S&P differences are smaller.

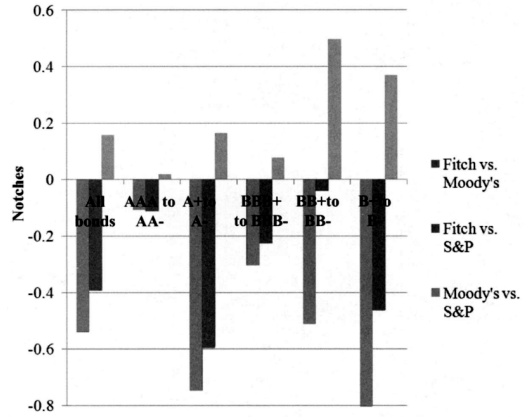
**Economic message:** Fitch's optimism is consistent with both rating-shopping and certification stories, so the rest of the paper asks whether markets treat Fitch as informative or as a regulatory tiebreaker.

**Table III**  
**Average Rating Differences**

Average rating differences for issues simultaneously rated by multiple CRAs, measured in rating notches, and split up by rating categories. Rating categories are defined by average Moody's and S&P ratings. We follow convention and use the numerical rating scale to convert ratings. For Fitch and S&P (with Moody's rating in parentheses), the numerical scores corresponding to the rating notches are, respectively, 1 for AAA (Aaa), 2 for AA+ (Aa1), 3 for AA (Aa2), 4 for AA- (Aa3), 5 for A+ (A1), 6 for A (A2), 7 for A- (A3), 8 for BBB+ (Baa1), 9 for BBB (Baa2), 10 for BBB- (Baa3), 11 for BB+ (Ba1), 12 for BB (Ba2), 13 for BB- (Ba3), 14 for B+ (B1), 15 for B (B2), and 16 for B- (B3). Therefore, a negative number means that the first-mentioned rating agency gives on average a better rating than the other CRA in that comparison. Quarterly data for 2000 to 2008 are used. *t*-statistics based on robust standard errors clustered by issuer are in brackets. \*, \*\*, and \*\*\* indicate statistical significance at the 10%, 5%, and 1% level, respectively.

|                  | Fitch vs.<br>Moody's | Fitch vs.<br>S&P     | Moody's vs. S&P<br>(Fitch Rated Sample) | Moody's vs. S&P<br>(Full Sample) |
|------------------|----------------------|----------------------|-----------------------------------------|----------------------------------|
| All Bonds        |                      |                      |                                         |                                  |
| Difference       | -0.538***<br>[-5.57] | -0.397***<br>[-3.99] | 0.154***<br>[2.79]                      | 0.185***<br>[4.41]               |
| <i>N</i>         | 70,624               | 70,747               | 70,738                                  | 104,435                          |
| <i>N</i> Issuers | 450                  | 449                  | 452                                     | 818                              |
| AAA to AA-       |                      |                      |                                         |                                  |
| Difference       | -0.111<br>[-1.53]    | -0.118<br>[-0.98]    | 0.0217<br>[0.14]                        | 0.0415<br>[0.47]                 |
| <i>N</i>         | 3,557                | 3,549                | 3,601                                   | 8,619                            |
| <i>N</i> Issuers | 37                   | 37                   | 37                                      | 57                               |
| A+ to A-         |                      |                      |                                         |                                  |
| Difference       | -0.751***<br>[-5.85] | -0.599***<br>[-4.63] | 0.165*<br>[1.80]                        | 0.203***<br>[2.68]               |
| <i>N</i>         | 36,461               | 36,539               | 36,532                                  | 48,905                           |
| <i>N</i> Issuers | 232                  | 232                  | 233                                     | 475                              |
| BBB+ to BBB-     |                      |                      |                                         |                                  |
| Difference       | -0.294***<br>[-5.03] | -0.223***<br>[-3.82] | 0.072<br>[1.11]                         | 0.117**<br>[2.42]                |
| <i>N</i>         | 24,177               | 24,193               | 24,186                                  | 36,841                           |
| <i>N</i> Issuers | 277                  | 278                  | 276                                     | 451                              |
| BB+ to BB-       |                      |                      |                                         |                                  |
| Difference       | -0.492***<br>[-4.76] | -0.0574<br>[-0.35]   | 0.478***<br>[3.30]                      | 0.471***<br>[4.71]               |
| <i>N</i>         | 6,429                | 6,466                | 6,419                                   | 10,070                           |
| <i>N</i> Issuers | 165                  | 163                  | 165                                     | 295                              |
| B+ to B-         |                      |                      |                                         |                                  |
| Difference       | -0.800***<br>[-6.80] | -0.457**<br>[-2.46]  | 0.369*<br>[1.82]                        | 0.430***<br>[3.20]               |
| <i>N</i>         | 3,279                | 3,272                | 3,282                                   | 6,200                            |
| <i>N</i> Issuers | 98                   | 98                   | 99                                      | 268                              |

(a) Table III: Average Rating Differences



(b) Figure 3: Rating Differences across CRAs

## 7.4 Table IV: Change in Credit Spreads and Rating Changes, Parts 1 and 2

Read:

- Table IV compares spread reactions to upgrades and downgrades from Moody's, S&P, and Fitch.
- Moody's and S&P downgrades are strongly associated with widening credit spreads.
- Fitch rating changes are not broadly informative in the same way once the other two agencies are included.
- The key exception is boundary certification: when Fitch changes the IG/HY classification, the spread effect is economically large.
- The table therefore rejects a broad information effect for Fitch but supports a boundary-specific certification effect.

**Economic message:** Bond investors do not appear to price Fitch revisions as general credit news; they price Fitch when it changes the regulatory class of the bond.

**Table IV**  
**Change in Credit Spreads and Rating Changes**

Using quarterly panel data between 2002 and 2008, we regress changes in credit spreads for bonds rated AAA to B— that are rated by all three CRAs on rating up- and downgrades for all three CRAs, changes in bond and firm characteristics, dummies for boundary effects and time fixed effects. Up- and downgrades are coded as dummies indicating whether each of the three CRAs upgraded or downgraded its rating. The following firm and bond controls are included but not shown (all in changes): leverage, liquidation/intrinsic value (PPE/total assets), R&D expenses (divided by total assets), ROA (return on assets), daily idiosyncratic equity volatility, historical equity beta (half-year daily corrected for bid-ask-bounces), log total assets (firm size, book value) and log offering size (issue size), redeemable (dummy for callability), duration and convexity. *Fitch Upgrade, Breaks Tie* is a dummy indicating that a Fitch upgrade made the issue qualify for IG, while *Fitch Downgrade, Breaks Tie* is a dummy indicating that a Fitch downgrade made the issue lose its IG qualification. *Fitch Could Break Tie* is a dummy indicating that the S&P and Moody's ratings are on opposite sides of the HY-IG boundary. Column (5) is restricted to issues rated A— or better by Moody's and S&P, whereas column (6) is restricted to issues rated BBB+ or worse by Moody's and S&P. *t*-statistics are in brackets (using robust standard errors clustered by issuer, *N*, issuer gives the number of issuers). \*, \*\*, and \*\*\* indicate significance at the 10%, 5%, and 1% level, respectively. *F-test Fup = Fdown* (*p*-value) gives the *p*-value for the coefficients on *Fitch Upgrade* and *Fitch Downgrade* being equal, while *F-test Fup, tie = Fdown, tie* (*p*-value) gives the *p*-value for the *F*-test of the coefficients on *Fitch Upgraded, Breaks Tie*, and *Fitch Downgraded, Breaks Tie* being equal.

|                           | (1)                 | (2)                 | (3)                 | (4)                | (5)               | (6)                  | (7)                | (8)                |
|---------------------------|---------------------|---------------------|---------------------|--------------------|-------------------|----------------------|--------------------|--------------------|
| Moody's Upgrade           | -4.501**<br>[-1.97] |                     |                     |                    |                   |                      | -2.369<br>[-1.11]  | -3.125<br>[-1.47]  |
| Moody's Downgrade         | 14.19***<br>[4.00]  |                     |                     |                    |                   |                      | 7.385*<br>[1.96]   | 7.575**<br>[2.10]  |
| S&P Upgrade               |                     | -5.594**<br>[-2.07] |                     |                    |                   |                      | -4.312<br>[-1.62]  | -4.423<br>[-1.59]  |
| S&P Downgrade             |                     | 22.34***<br>[5.25]  |                     |                    |                   |                      | 18.75***<br>[4.26] | 17.32***<br>[4.43] |
| Fitch Upgrade             |                     |                     | -6.463**<br>[-2.10] | -5.612*<br>[-1.83] | 6.975<br>[0.59]   | -8.209***<br>[-2.70] | -4.807<br>[-1.59]  | -3.892<br>[-1.30]  |
| Fitch Downgrade           |                     |                     | 12.98**<br>[2.46]   | 10.52**<br>[1.99]  | -0.595<br>[-0.26] | 22.30***<br>[4.00]   | 4.596<br>[0.80]    | 2.875<br>[0.49]    |
| Fitch Upgrade, Breaks Tie |                     |                     |                     | -17.59<br>[-1.18]  |                   | -16.26<br>[-1.11]    |                    | -14.94<br>[-1.03]  |

(continued)

**Table IV—Continued**

|                                                       | (1)                   | (2)                   | (3)                   | (4)                   | (5)                   | (6)                   | (7)                   | (8)                   |
|-------------------------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Fitch Downgrade, Breaks Tie                           |                       |                       |                       | 31.53*<br>[1.76]      |                       | 24.67<br>[1.45]       |                       | 26.65<br>[1.49]       |
| Fitch Could Break Tie                                 |                       |                       |                       | 13.81**<br>[2.39]     |                       | 9.475*<br>[1.75]      |                       | 13.33**<br>[2.48]     |
| Lagged Credit Spread Change                           | -0.253***<br>[-11.07] | -0.258***<br>[-11.71] | -0.251***<br>[-11.08] | -0.253***<br>[-11.48] | -0.400***<br>[-15.02] | -0.255***<br>[-10.72] | -0.259***<br>[-11.72] | -0.261***<br>[-12.01] |
| Sample                                                | All                   | All                   | All                   | All                   | ≤A—                   | ≥BBB+                 | All                   | All                   |
| <i>N</i>                                              | 24,282                | 24,282                | 24,282                | 24,282                | 11,330                | 12,952                | 24,282                | 24,282                |
| Adj. <i>R</i> <sup>2</sup>                            | 0.531                 | 0.533                 | 0.531                 | 0.533                 | 0.45                  | 0.605                 | 0.534                 | 0.536                 |
| <i>N</i> Issuers                                      | 380                   | 380                   | 380                   | 380                   | 117                   | 313                   | 380                   | 380                   |
| <i>F</i> -test Fup=Fdown ( <i>p</i> -value)           |                       |                       | 0.15%                 |                       | 56.30%                | 0.00%                 | 15.00%                | 30.80%                |
| <i>F</i> -test Fup, tie=Fdown, tie ( <i>p</i> -value) |                       |                       |                       | 2.82%                 |                       | 5.87%                 |                       | 6.07%                 |

(a) Table IV: Change in Credit Spreads and Rating Changes, Part 1

(b) Table IV: Change in Credit Spreads and Rating Changes, Part 2

## 7.5 Table V: Credit Spread Regressions on Fitch Rating Additions, Parts 1 and 2

Read:

- Table V focuses on the addition of Fitch after a bond has already been in the sample without Fitch.
- A Fitch addition by itself is not associated with a strong unconditional spread decline.
- The economically meaningful effect appears when Fitch makes the bond investment grade at the HY-IG boundary.
- Fitch additions that are equal to or worse than the existing ratings do not generate comparable spread improvements.
- The results are inconsistent with broad information production or simple market-believed rating shopping.

**Economic message:** The value of a third rating depends on the regulatory classification it produces, not simply on the existence of another opinion.

**Table V**  
**Credit Spread Regressions on Fitch Rating Additions**

Using quarterly panel data between 2002 and 2008, we regress changes in credit spreads for AAA to B- rated bonds that are rated by Moody's and S&P on rating additions from Fitch, the relative ranking of those additions, whether additions happened at the HY-IG boundary, interactions with uncertainty measures, changes in bond and firm characteristics, and time fixed effects. *Fitch Added*, *Better*, *Fitch Added, Equal*, and *Fitch Added, Worse* are dummies indicating whether a Fitch rating has been added that is, respectively, better than, equal to, and worse than the average rating by Moody's and S&P. *Fitch Added, Makes IG* and *Fitch Added, Denies IG* are dummies that indicate whether the added Fitch rating makes the issue qualify for IG or not, conditional on Moody's and S&P ratings being on opposite sides of the boundary. See Table IV for descriptions of bond- and firm-level control variables. *t*-statistics are in brackets (using robust standard errors clustered by issuer in all columns except column 5, which uses double clustering by both issuer and time). \*, \*\*, and \*\*\* indicate significance at the 10%, 5%, and 1% level, respectively. *F*-test *Fadded*, *IG = Fadded*, *HY* (*p*-value) is the *p*-value of the *F*-test of *Fitch Added*, *Makes IG* and *Fitch Added*, *Makes HY* being equal. The sample comprises all issues rated by both Moody's and S&P with their average rating better or equal to B-, except in column 4, where their average rating is between BBB+ and BB-.

|                       | (1)              | (2)               | (3)                 | (4)                 | (5)                 | (6)                | (7)                   |
|-----------------------|------------------|-------------------|---------------------|---------------------|---------------------|--------------------|-----------------------|
| Fitch Added           | -2.28<br>[-0.65] |                   |                     |                     |                     |                    |                       |
| Fitch Added, Better   |                  | -5.832<br>[-0.83] | -5.254<br>[-0.75]   | -8.644<br>[-1.16]   | -5.254<br>[-0.66]   | -7.243<br>[-1.00]  | -6.197<br>[-0.88]     |
| Fitch Added, Equal    |                  | -0.205<br>[-0.06] | 0.0847<br>[0.03]    | 3.228<br>[0.46]     | 0.0847<br>[0.02]    | 2.14<br>[0.59]     | 3.098<br>[0.51]       |
| Fitch Added, Worse    |                  | 4.422<br>[0.94]   | 5.089<br>[0.94]     | 6.347<br>[0.81]     | 5.089<br>[0.98]     | 3.063<br>[0.40]    | 5.383<br>[0.97]       |
| Fitch Added, Makes IG |                  |                   | -30.47**<br>[-2.54] | -23.00**<br>[-1.98] | -30.47**<br>[-2.76] |                    |                       |
| Fitch Added, Makes HY |                  |                   |                     | 22.99<br>[0.65]     | 10.8<br>[0.57]      |                    |                       |
| At HY-IG Boundary     |                  |                   | 15.32***<br>[3.22]  | 10.93**<br>[2.66]   | 15.32**<br>[2.23]   |                    |                       |
| Fitch Added, Equal*   |                  |                   |                     |                     |                     | -949.1<br>[-0.90]  | 2.417**<br>[2.02]     |
| Analyst Dispersion    |                  |                   |                     |                     |                     | 700.4***<br>[6.24] | -0.258***<br>[-14.67] |
| Fitch Added, Equal*   |                  |                   |                     |                     |                     |                    | 0.531<br>[0.82]       |
| Rating Dispersion     |                  |                   |                     |                     |                     |                    | 0.531<br>[0.82]       |

*(continued)*

**Table V—Continued**

|                                                                                  | (1) | (2)                   | (3)                   | (4)                   | (5)                   | (6)                  | (7)                   |
|----------------------------------------------------------------------------------|-----|-----------------------|-----------------------|-----------------------|-----------------------|----------------------|-----------------------|
| S&P and Moody's                                                                  |     |                       |                       |                       |                       |                      |                       |
| Diagrees                                                                         |     |                       |                       |                       |                       |                      | 2.417**<br>[2.02]     |
| Lagged Credit Spread                                                             |     | -0.257***<br>[-14.39] | -0.257***<br>[-14.40] | -0.258***<br>[-14.81] | -0.277***<br>[-12.78] | -0.258***<br>[-5.78] | -0.265***<br>[-14.96] |
| Change                                                                           |     | All                   | All                   | All                   | BBB+ to BB-           | All                  | All                   |
| Sample                                                                           |     | No                    | No                    | No                    | No                    | No                   | No                    |
| Double Clustering                                                                |     | No                    | No                    | No                    | No                    | No                   | No                    |
| <i>N</i>                                                                         |     | 34,568                | 34,568                | 34,568                | 15,058                | 34,568               | 34,568                |
| Adj. <i>R</i> <sup>2</sup>                                                       |     | 0.531                 | 0.531                 | 0.532                 | 0.628                 | 0.532                | 0.535                 |
| <i>N</i> Issuers                                                                 |     | 668                   | 668                   | 668                   | 463                   | 668                  | 668                   |
| <i>F</i> -test <i>Fadded</i> , <i>IG = Fadded</i> , <i>HY</i> ( <i>p</i> -value) |     |                       |                       | 3.23%                 | 5.58%                 | 9.35%                |                       |

(a) Table V: Credit Spread Regressions on Fitch Rating Additions, Part 1

(b) Table V: Credit Spread Regressions on Fitch Rating Additions, Part 2

## 7.6 Table VI: Cox Regressions for Time to Adding Fitch Rating + Figure 4: Default Prediction Accuracy

Read:

- Table VI models the timing of obtaining a Fitch rating. The event is getting Fitch coverage.
- The strongest selection effect is the *Fitch Could Push IG* or tie-breaking variable: issues are more likely to add Fitch when it can move them into IG status.
- By contrast, uncertainty variables such as analyst dispersion and idiosyncratic volatility do not consistently predict adding Fitch in the direction required by information production.
- Figure 4 compares default-prediction accuracy. Moody's and S&P perform at least as well as, and generally better than, Fitch.
- Fitch does not add strong default-prediction information once the other ratings are considered.

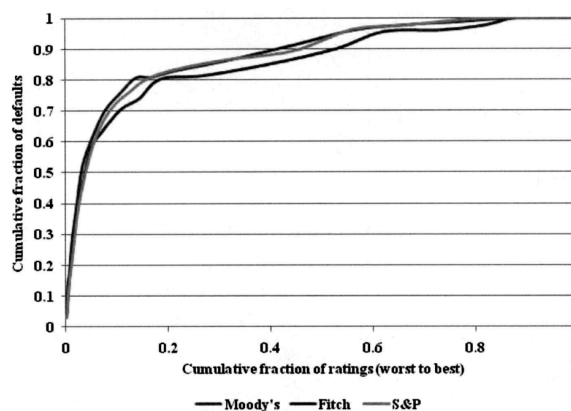
**Economic message:** Issuers appear to seek Fitch when it has regulatory value, and Fitch ratings do not dominate in predicting default risk.

**Table VI**  
**Cox Regressions for Time to Adding Fitch Rating**

Cox proportional hazard model regressions of the time to adding a Fitch rating on rating category dummies based on average Moody's and S&P (MSP) ratings, measures of uncertainty *Analyst Dispersion* (standard deviation of analyst earning forecasts normalized per dollar share value) and *MSP Rating Dispersion* (absolute value of the notches difference between Moody's and S&P), and whether the Fitch rating "could push" the issue to IG or A-. *F Could Push IG* and *F Could Push A-* are dummies indicating whether the Moody's and S&P ratings are on opposite sides of the IG or A- boundary, respectively. Other controls that are included but not shown are firm beta, leverage, PPE/assets, R&D expenses/assets, ROA, log of offering amount, time to maturity, time to maturity squared and redeemable (see Table IV for descriptions). Quarterly data for 2000 to 2008 are used. The sample consists of all issues with both Moody's and S&P ratings that on average are rated B- or better. Coefficients on the covariates in the partial hazard function are reported, and *t*-statistics are in brackets (using robust standard errors clustered by issuer). \*, \*\*, and \*\*\* indicate statistical significance at the 10%, 5%, and 1% level, respectively. Pseudo  $R^2$  refers to McFadden (1973) pseudo  $R^2$ .

|                          | (1)                  | (2)                  | (3)                  | (4)                  | (5)                  |
|--------------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
| MSP A+ to A- Rating      | 1.553***<br>[3.04]   |                      | 1.548***<br>[3.06]   | 1.568***<br>[3.08]   | 1.524***<br>[3.06]   |
| MSP BBB+ to BBB- Rating  | 1.146**<br>[2.28]    |                      | 1.156**<br>[2.32]    | 1.119**<br>[2.22]    | 1.071**<br>[2.19]    |
| MSP BB+ to BB- Rating    | 1.346**<br>[2.53]    |                      | 1.481***<br>[2.79]   | 1.408***<br>[2.60]   | 1.292**<br>[2.49]    |
| MSP B+ to B- Rating      | 1.131**<br>[2.06]    |                      | 1.114**<br>[2.06]    | 1.159**<br>[2.12]    | 1.073**<br>[2.00]    |
| Fitch Could Push IG      | 0.717***<br>[2.83]   | 0.693***<br>[2.83]   |                      |                      | 0.702**<br>[2.44]    |
| Fitch Could Push A       | 0.248<br>[1.23]      | 0.221<br>[1.13]      |                      |                      | 0.231<br>[1.00]      |
| Avg MSP BB+              |                      |                      |                      | 0.292<br>[1.27]      |                      |
| Avg MSP BBB-             |                      |                      |                      | 0.235<br>[1.41]      |                      |
| MSP Rating Dispersion    |                      |                      |                      |                      | -0.128<br>[-0.72]    |
| S&P and Moody's Disagree | -0.307<br>[-1.16]    | -0.267<br>[-0.90]    | -0.254<br>[-1.03]    | -0.251<br>[-1.02]    |                      |
| Analyst Dispersion       | -4.988<br>[-0.58]    | -4.262<br>[-0.53]    | -6.275<br>[-0.69]    | -6.479<br>[-0.70]    | -5.979<br>[-0.64]    |
| Idiosyncratic Volatility | -26.11***<br>[-3.95] | -23.85***<br>[-3.86] | -25.62***<br>[-3.86] | -26.08***<br>[-3.93] | -26.61***<br>[-3.88] |
| Log of Total Assets      | 0.296***<br>[5.53]   | 0.200**<br>[2.57]    | 0.290***<br>[5.39]   | 0.299***<br>[5.67]   | 0.289***<br>[5.26]   |
| <i>N</i>                 | 38,351               | 38,351               | 38,351               | 38,351               | 38,351               |
| Pseudo $R^2$             | 0.029                | 0.019                | 0.029                | 0.029                | 0.028                |
| <i>N</i> . Issuers       | 813                  | 813                  | 813                  | 813                  | 813                  |

(a) Table VI: Cox Regressions for Time to Adding Fitch Rating



(b) Figure 4: Default Prediction Accuracy

## 7.7 Table VII: CRA Default Prediction, Parts 1 and 2

Read:

- Panel A of Table VII reports logit regressions of default on rating measures.
- Moody's and S&P ratings have strong predictive power for default.
- Fitch ratings perform worse and do not add much when Moody's and S&P ratings are already in the specification.
- Panel B reports accuracy ratios over one- and two-year horizons. Moody's has the highest accuracy, followed by S&P and then Fitch.

**Economic message:** The default-prediction evidence weakens the information-production hypothesis for the marginal Fitch rating.

**Table VII**  
**CRA Default Prediction**

Using yearly panel data between 2000 and 2008, we compare CRA performance with respect to default prediction on a 1-year horizon. Panel A shows results from a logit regression of issuer default events on rating scales, rating scale dummies, and rating differences for the complete universe of bonds that are rated by all three CRAs. The rating variables are in notches, where 1 corresponds to AAA, 2 to AAA-, etc. The difference variables measure the difference between the first minus the second CRA in notches. Only marginal effects are reported (multiplied by 10,000), and *t*-statistics are in brackets (using robust standard errors clustered by issuer). Panel B shows accuracy ratios and their differences for all three CRAs for a 1- and 2-year forecasting horizon. Standard errors are constructed using the Jackknife method with resampling at the issuer level (equivalent to clustering by issuer). \*, \*\*, and \*\*\* indicate significance at the 10%, 5%, and 1% level, respectively.

|                                      | (1)     | (2)               | (3)               | (4)     | (5)               | (6)               | (7)               | (8)                | (9)               | (10)              | (11)              | (12)              |
|--------------------------------------|---------|-------------------|-------------------|---------|-------------------|-------------------|-------------------|--------------------|-------------------|-------------------|-------------------|-------------------|
| Moody's Rating                       |         | 4.23***<br>[4.60] |                   |         |                   |                   |                   | 04.25***<br>[4.59] |                   |                   |                   | 4.22***<br>[4.55] |
| S&P Rating                           |         |                   | 5.02***<br>[4.98] |         |                   |                   |                   |                    | 5.08***<br>[4.97] | 4.22***<br>[4.55] |                   |                   |
| Fitch Rating                         |         |                   |                   |         | 5.98***<br>[5.30] | 4.25***<br>[4.59] |                   | 05.08***<br>[4.97] |                   |                   |                   |                   |
| Difference Moody's and Fitch Ratings |         |                   |                   |         |                   | 3.94***<br>[4.27] | -0.306<br>[-0.75] |                    |                   |                   |                   |                   |
| Difference S&P and Fitch Ratings     |         |                   |                   |         |                   |                   |                   | 3.84***<br>[3.72]  | -1.24<br>[-1.49]  |                   |                   |                   |
| Difference Moody's and S&P Ratings   |         |                   |                   |         |                   |                   |                   |                    |                   | 3.57***<br>[4.96] | -0.646<br>[-1.40] |                   |
| Moody's Rat. FE                      | Yes     | No                | No                | No      | No                | No                | No                | No                 | No                | No                | No                | No                |
| Fitch Rat. FE                        | No      | No                | No                | No      | Yes               | No                | No                | No                 | No                | No                | No                | No                |
| S&P Rat. FE                          | No      | No                | Yes               | No      | No                | No                | No                | No                 | No                | No                | No                | No                |
| <i>N</i>                             | 104,966 | 112,329           | 97,371            | 112,329 | 103,803           | 112,329           | 112,329           | 112,329            | 112,329           | 112,329           | 112,329           | 112,329           |
| Pseudo <i>R</i> <sup>2</sup>         | 0.371   | 0.366             | 0.333             | 0.333   | 0.323             | 0.308             | 0.367             | 0.367              | 0.337             | 0.337             | 0.367             | 0.367             |
| <i>N</i> Issuers                     | 1,994   | 2,066             | 1,949             | 2,066   | 1,984             | 2,066             | 2,066             | 2,066              | 2,066             | 2,066             | 2,066             | 2,066             |

*(continued)*

(a) Table VII: CRA Default Prediction, Panel A

**Table VII—Continued**

Panel B: Accuracy Ratios

|                      | 1-year horizon |                | 2-year horizon |                |
|----------------------|----------------|----------------|----------------|----------------|
|                      | Value          | <i>t</i> -stat | Value          | <i>t</i> -stat |
| Moody's Rating       | 0.779***       | 25.38          | 0.713***       | 21.05          |
| Fitch Rating         | 0.718***       | 18.5           | 0.657***       | 15.97          |
| S&P Rating           | 0.765***       | 24.12          | 0.704***       | 20.15          |
| Moody's–Fitch Rating | 0.062**        | 2.46           | 0.057**        | 2.25           |
| S&P–Fitch Rating     | 0.047*         | 1.83           | 0.047**        | 1.97           |
| Moody's–S&P Rating   | 0.015          | 1.31           | 0.009          | 0.8            |

(b) Table VII: CRA Default Prediction, Panel B

## 7.8 Table VIII: Relative Rating Levels + Table IX: Logistic Regressions for Having a Rating Transition

Read:

- Table VIII asks whether the additional Fitch rating leads to a regulatory gain conditional on boundary positioning.
- *Fitch Could Push IG* is positive and significant, meaning Fitch is more likely to generate a regulatory improvement when it can break the HY–IG tie.
- Table IX shows that having Fitch is associated with a higher probability of future rating transitions by Moody's and S&P.
- This pattern supports a precautionary motive: issuers whose ratings are likely to move have reason to obtain a third rating in advance.

**Economic message:** The marginal rating is valuable not only today but also as insurance against future transitions that could otherwise push a bond below investment grade.

Table VIII  
Relative Rating Levels

The table shows logit regressions of achieving a regulatory gain on rating category dummies, *Analyst Dispersion* (standard deviation of analyst earning forecasts normalized per dollar share value) as a measure of uncertainty, and whether the Fitch rating “could push” the issue to IG. *F Could Push IG* is a dummy indicating whether Moody’s and S&P ratings are on opposite sides of the HY–IG boundary. Other controls that are included but not shown are firm beta, leverage, PPE/assets, R&D expenses/assets, ROA, log of offering amount, time to maturity, time to maturity squared, and redeemable (see Table IV for descriptions). Quarterly data for 2000 to 2008 are used. The sample consists of all issues rated by Moody’s, S&P, and Fitch that on average are rated B– or better and for which Moody’s and S&P disagree. Marginal effects are reported for the regressions and all *t*-statistics are in brackets (using robust standard errors clustered by issuer). \*, \*\*, and \*\*\* indicate statistical significance at the 10%, 5%, and 1% level, respectively.

|                              | (1)                | (2)                | (3)               |
|------------------------------|--------------------|--------------------|-------------------|
| A+ to A–                     | 0.0691<br>[0.43]   |                    | 0.0698<br>[0.42]  |
| BBB+ to BBB–                 | –0.155<br>[–0.76]  |                    | –0.155<br>[–0.75] |
| BB+ to BB–                   | –0.0823<br>[–0.34] |                    | –0.016<br>[–0.08] |
| B+ to B–                     | –0.0516<br>[–0.19] |                    | –0.052<br>[–0.19] |
| Fitch Could Push IG          | 0.187***<br>[3.73] | 0.196***<br>[3.62] |                   |
| Analyst Dispersion           | 5.461<br>[1.44]    | 5.366<br>[1.60]    | 5.192<br>[1.41]   |
| Idiosyncratic Volatility     | –1.432<br>[–0.34]  | –2.132<br>[–0.44]  | –1.735<br>[–0.40] |
| Log of Total Assets          | 0.0351<br>[0.90]   | 0.0476<br>[1.41]   | 0.0316<br>[0.80]  |
| Other Controls Included      | Yes                | Yes                | Yes               |
| Only with MSP Disagreement   | Yes                | Yes                | Yes               |
| Fitch Added                  | Yes                | Yes                | Yes               |
| Industry FE                  | Yes                | Yes                | Yes               |
| <i>N</i>                     | 569                | 569                | 569               |
| Pseudo <i>R</i> <sup>2</sup> | 0.15               | 0.114              | 0.131             |
| <i>N</i> . Issuers           | 170                | 170                | 170               |

(a) Table VIII: Relative Rating Levels

Table IX  
Logistic Regressions for Having a Rating Transition

Logit regressions of having a rating transition next quarter on rating category dummies (AAA and AA+ are merged to avoid singularities), a dummy indicating whether the issue has a Fitch rating, measures of uncertainty as idiosyncratic volatility, beta, *Analyst Dispersion* (standard deviation of analyst earning forecasts normalized per dollar share value) and *MSP Rating Dispersion* (absolute value of the notches difference between Moody’s and S&P). See Table IV for descriptions of bond- and firm-level control variables. Quarterly data for 2000 to 2008 are used. The sample consists of all issues with both Moody’s and S&P ratings. Only marginal effects are reported, and *t*-statistics are in brackets (using robust standard errors clustered by issuer). \*, \*\*, and \*\*\* indicate statistical significance at the 10%, 5%, and 1% level, respectively.

|                              | Moody’s Rating Change | S&P Rating Change   |
|------------------------------|-----------------------|---------------------|
| Fitch Rated                  | 0.0128***<br>[3.68]   | 0.0102***<br>[2.94] |
| Idiosyncratic Volatility     | 1.509***<br>[5.95]    | 0.712***<br>[3.17]  |
| Beta                         | –0.00562<br>[–1.21]   | 0.00528<br>[0.96]   |
| Analyst Dispersion           | 0.0459<br>[1.27]      | 0.0549<br>[1.51]    |
| MSP Rating Dispersion        | 0.0128***<br>[4.55]   | 0.00611**<br>[2.16] |
| MSP Rating FE                | Yes                   | Yes                 |
| Year FE                      | Yes                   | Yes                 |
| <i>N</i>                     | 104,629               | 104,629             |
| Pseudo <i>R</i> <sup>2</sup> | 0.082                 | 0.067               |
| <i>N</i> . Issuers           | 818                   | 818                 |

regulatory gain is impossible. The table provides suggestive evidence that the (additional) Fitch rating is more optimistic around the HY–IG boundary and

(b) Table IX: Logistic Regressions for Having a Rating Transition

## 7.9 Table X: Turnover Regressions

Read:

- Table X links certification to liquidity.
- Bonds in lower rating categories generally have lower turnover.
- If Fitch makes the bond investment grade, turnover improves, consistent with entry into the eligible investment universe of regulated or mandate-constrained investors.
- If Fitch denies IG at the boundary, liquidity is substantially worse.
- The turnover evidence reinforces the view that Fitch affects market segmentation, not merely beliefs about default.

**Economic message:** Certification changes who can hold and trade the bond; the liquidity margin is a crucial part of the value of the third rating.

**Table X**  
**Turnover Regressions**

OLS regressions of quarterly bond turnover, measured as aggregated trading volume over total value of outstanding bonds on rating category dummies, a dummy indicating whether the issue has a Fitch rating, and controls for off-the-run vs. on-the-run effects (*Age*). *F Makes (Denies) IG* is a dummy equal to one if Moody's and S&P ratings are on opposite sides of the HY-IG boundary and the Fitch rating is IG (HY). All other control variables are dropped due to the use of both time and issue fixed effects. Monthly data for July 2002 to December 2008 are used. The sample consists of all issues with both Moody's and S&P ratings. *t*-statistics are in brackets (using robust standard errors clustered by issuer). \*, \*\*, and \*\*\* indicate statistical significance at the 10%, 5%, and 1% level, respectively.

|                            | (1)                  | (2)                  | (3)                  |
|----------------------------|----------------------|----------------------|----------------------|
| MSP AAA to AA- Rating      | -2.889***<br>[-2.92] | -2.887***<br>[-2.91] | -2.891***<br>[-2.91] |
| MSP BBB+ to BBB- Rating    | 0.722<br>[0.97]      | 0.749<br>[1.00]      | 0.745<br>[0.94]      |
| MSP BB+ to BB- Rating      | -2.434*<br>[-1.83]   | -2.543*<br>[-1.78]   | -2.554*<br>[-1.77]   |
| MSP B+ to B- Rating        | -5.368***<br>[-3.51] | -5.270***<br>[-3.40] | -5.280***<br>[-3.38] |
| Age                        | -4.035***<br>[-6.91] | -4.033***<br>[-6.92] | -4.022***<br>[-6.91] |
| F Makes IG                 |                      | 4.208***<br>[2.67]   | 4.213***<br>[2.67]   |
| F Denies IG                |                      | -5.801**<br>[-2.18]  | -5.767**<br>[-2.16]  |
| Fitch Rated                |                      |                      | 0.952<br>[1.40]      |
| Time FE                    | Yes                  | Yes                  | Yes                  |
| Issue FE                   | Yes                  | Yes                  | Yes                  |
| <i>N</i>                   | 43,292               | 43,292               | 43,292               |
| Adj. <i>R</i> <sup>2</sup> | 0.278                | 0.279                | 0.279                |
| N. Issuers                 | 739                  | 739                  | 739                  |

Table X: Turnover Regressions

## 8 Short summary

- The paper studies why corporate bonds have multiple credit ratings.
- Almost all large bonds have Moody's and S&P ratings, so Fitch is interpreted as the marginal third rating.
- The paper evaluates information production, rating shopping, and regulatory certification.
- Fitch is generally more optimistic than Moody's and S&P.
- Fitch does not generally move spreads or improve default prediction after controlling for Moody's and S&P.
- Fitch matters strongly when it can change the bond's IG/HY classification.
- The probability of adding Fitch is higher when Moody's and S&P straddle the HY-IG boundary.
- A Fitch rating that makes the bond investment grade is associated with lower spreads and better liquidity.
- Default prediction tests show Moody's and S&P dominate Fitch.
- The core conclusion is that Fitch acts mainly as a regulatory certification tiebreaker rather than as a broad producer of new credit information.

## 9 Conclusion

### 9.1 Core conclusion

Bongaerts, Cremers, and Goetzmann show that the marginal value of an additional credit rating is shaped by regulation and investor mandates. Fitch ratings are not broadly priced as new credit information after



Moody's and S&P are controlled for. Instead, Fitch matters primarily when it changes a bond's regulatory status from high yield to investment grade.

## **9.2 Economic intuition**

The intuition is built around a rating rule. With two ratings, investors and regulators often use the lower rating. With three ratings, they use the middle rating. When Moody's and S&P disagree across the HY–IG boundary, a third Fitch rating becomes decisive. A favorable Fitch rating can convert a bond into an investment-grade security, expanding its potential investor base, lowering spreads, and increasing liquidity. An unfavorable Fitch rating, by contrast, does not provide comparable benefits.

## **9.3 Incremental contribution revisited**

The paper's incremental contribution is to demonstrate empirically that credit ratings are not merely beliefs about default probabilities. They are also institutional keys that determine eligibility for regulated investors, index inclusion, contract triggers, and safe-harbor classifications. This moves the literature beyond the question of whether ratings are informative and toward a richer view: ratings can be privately valuable because they coordinate market segmentation.

## **9.4 Implications for rating-agency competition**

The paper also changes how one should think about competition among rating agencies. A third agency can survive not because it produces the best default model, but because its ratings interact with rules that use multiple ratings mechanically. If regulation gives the third rating a special role, an optimistic marginal CRA can provide value even if its information content is limited. This creates incentives that differ from those in a pure information market.

## **9.5 Policy implications**

The evidence implies that reducing mechanical reliance on ratings could change the demand for multiple ratings. If rules relied less on median ratings or bright-line IG/HY thresholds, issuers would have less incentive to seek a third rating for certification reasons. However, if ratings remain embedded in regulation and mandates, the value of ratings will continue to include both information and eligibility effects.

## **9.6 Limitations and open questions**

The empirical design is powerful but not randomized. Issuers choose whether to obtain Fitch, and the decision is likely related to unobserved issuer incentives. The paper addresses this using event timing, controls, default prediction, and liquidity evidence, but residual selection may remain. Another limitation is that the sample period ends in 2008; post-crisis reforms and changing regulatory reliance on ratings could alter the strength of the certification channel.

## **9.7 Takeaway**

The enduring lesson is that information intermediaries can be valuable because of the institutional rules that define how their signals are used. In this market, the third credit rating is valuable mainly as a tiebreaker at a regulatory boundary. The paper therefore provides a sharp example of how regulation can shape the production, demand, and pricing of financial information.